

A Scalable Architecture for Rapid Development of Trackers amid Pandemics like COVID-19

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Abstract—Pandemics in history caused enormous negative impacts on health, economy, social and lives across the globe. There have been many such recorded medical outbreaks including Spanish Flu, SARS, Ebola, Zika, and recently reported COVID 19. Any such pandemics are stretching far beyond the spread of the disease now. Researchers claimed that most of the reported live loss cases are due to a lack of medical facilities available in the region. While officials and health care officers promise to deliver timely medical services but still there exists a gap in cascading the information from servicemen to commoners. This generates the requirement of a new development delivery model that is rapid, efficient, and cost-effective. In this paper, we are proposing an architecture based on the client-server model that can handle the dynamic volume, velocity, and variety of information amid such a crisis. This architecture can be used for showcasing history data as well as to provide real-time information about the health care facility including hospitals, pathology labs, medical equipment's, facilities with user-driven route locator system for reaching out best possible medical care units within a stipulated time limit. To showcase the effectiveness of the proposed architecture, a prototype tracker has been developed and deployed for COVID-19 with functionalities like available required medical facilities. This can be drilled down by the user across the country, state, city, region, location.

Keywords—Architecture, Framework, Pandemics, Tracker, Medical Resource Tracker, COVID-19, Data Visualization, Data Analysis

I. INTRODUCTION

In the realm of infectious diseases, a pandemic is a worst-case scenario. A disease or a medical emergency can be termed a pandemic if its spread crosses any country's boundary. In the current global scenario, this situation is quite practical as humans are traveling around the world on a day-to-day basis. From 430 BC till today many such incidents have been recorded where pandemics hit the global population. This recurrent crisis has disturbed quadrillions of living and taken away billions of lives from time to time. The exemplar of such an emergency that is persisting and prevailing worldwide currently is COVID-19[1]. This is a disease caused by the "novel coronavirus" with general symptoms like fever, dry cough, breathing difficulty, and less common symptoms as aches, pain, sore throat, diarrhea, etc. [1]. Since, these symptoms range from mild to severe having major dependence on a person's age, gender, stress level as well as past medical history. A lack of adequate knowledge is probably the driving force for public panic. Amid this situation, individuals agonize from anxiety and depression due to fake or obsolete news bulletins regarding country medical facilities. Considering the prevalent situation, the cause and treatment of such new viruses are still unknown as there can

be very few perfectly defined symptoms respective to these diseases [2]. The only way to keep yourself well and unharmed is awareness. Staying aware of the latest information on these pandemics like COVID-19 is necessary so that one should not get panic and do not dread.

To spread the consciousness and relieve the public agitation, one forward step would be the timely publication of trustworthy research evidence based on the country's real-time medical facilities. These data need to be stored, managed, analyzed, and broadcasted among the locals rapidly. In a country like India, the voluminous medical data that can be gathered from global governmental medical bodies like WHO, IARC, UNICEF, MOHFW, etc. needs to accumulate, managed, and analyze to acquaint the local public.

As per World Economic Forum [3], the whole world is contributing around 2.5 quintillion bytes of data every day, and specifically, 90% of its part is created majorly within the last 02 years. The above-mentioned bodies release the medical data so frequently that traditional methods of data storage are not fruitful. If by any means we store this voluminous data, it has become difficult to manage and make sense of all of it on the go. It is near to impossible for a person to go through this huge data collected line-by-line and make out distinct patterns and observations. An architecture supporting the analysis of a complete dataset just by visualizing the same can be proven quite beneficial in the paradigm.

Data visualization [4] refers to the technique, in which visual objects and graphics are used to communicate data. The goal behind this is to maintain clarity in the information that is being shared. In the current era, visualizations play more hold on information than just representation. It helps in easy and accessible access to check and analyze the data set patterns, trends, outliers, etc. [6]. Since this field has the capability of providing insight that traditional descriptive statistics cannot, makes it more useful. In our prototype tracker development, we have used a means of visualization to display all the collected data from authentic sources related to COVID 19. So that the statistics can easily be understood by everyone who just has a little knowledge about the internet and how to work with it. Beyond communicating and analyzing the data, visualization also plays a role in helping to convince people to change their behavior by gaining real-time information about the spread of medical emergencies and real-time information about the country's health care system. Not the casual traditional data visualization but an interactive data visualization is in high demand to persuade and understand current conditions and to see the future aspect of the pandemic. Keeping a sharp view on monitoring the available medical facilities, country's medical situation, new requirements, equipment, revamping and refurbishment of labs can help minimize human life loss and control this

exigency [12]. Initially, we designed one client-server architecture namely, generic 04-layer architecture for monitoring and broadcasting useful information. This architecture was having various performances and was not supporting much quick delivery. This has been modified and an enhanced version of the same name, layer 03 architecture. The specification of each design is present in the proposed methodology section. Our contribution for resolving the above-discussed issues are as follows:

1. A generic 04-layer architecture for the development and deployment of any tracking, monitoring, or broadcasting system used during and after the pandemic.
2. An enhanced 03-layer architecture for rapid design, development, and deployment of varied generic trackers during and after the pandemic.
3. A developed prototype tracker for the available medical facility.
4. The prototype has strengthened its usage by providing accurate information about the geographical location of nearby hospitals, testing labs based on the user's GPS location.
5. The prototype is supported by a tracking and route locator system for quick medical assistance.
6. We further added required and valuable information pages like pandemics helpline numbers, frequently asked questionnaires by and for the people, Broadcast bulletins from local or global medical authorities in our prototype.

II. LITERATURE REVIEW

Over the years, Plagues and Pandemics have ravaged humanity throughout, sometimes leading to change the history itself and even seeing an end of civilizations. Computer Science and its application always help the world to fight during the time of the pandemic. Emerging technologies like AI and ML can brace the community by predicting the current as well as future preparedness for medical emergencies. In this section, we outline the related work carried in this area. This review is essential to the contrast between the proposed architecture and currently implemented work, including its advantages over the existing related monitoring systems.

Various trackers and monitoring system has been developed by competent researchers. Symptom Monitoring System for contact tracing is been exhibited in [2]. A technological solution for providing and tracing symptoms specific to SARS-COV-2 is emphasized in the given work. Further, the need for a drug and treatment tracker is described in [3]. Here the focus is majorly on the theoretical aspect of this tracker rather implementation of the same is still a gap. A hospital epidemics tracker is implemented by [4]. The main aim of their work is to monitor COVID-19 and other epidemics in hospital workers. The work has been expanded on the case and contact identification, namely test-trace-track-isolate within hospital staff, as they were becoming infected with COVID-19 disproportionately more frequently and severely than the general population. Another android-based Medicine tracking application is developed in [5]. It is intended for users who need a little assistance in keeping with following their drug plan. The collateral damage due to pandemics like COVID 19 is been discussed by senior surgical practioner in [6]. A case study of a 48 years adult is focused here and how drastically delay in medical care due to COVID 19, can affect a sound and healthy person is of prime concern. The shifting in the healthcare delivery mechanism is

explored in [7]. This work states the recent shift of general health services to e-health services during pandemics like COVID-19. Therefore, essential services like food, transport, and most importantly healthcare are of extreme importance, and making all of these digitally available can boost any country's preparedness plan during or after pandemics.

On-time availability of each of the required resources is a matter of concern and the information is required to be dispersed among the needy within the time is a prime goal during any pandemic. This issue has been enlightened in [8], where the toughest triage of allocating and meeting the required need of these healthcare units is showcased.

Looking at previous work in this domain we can readily conclude a few points. One, that maximum of the work has been specific to COVID-19 rather than pandemics. Secondly, the need for amalgaming medical services with technology [9] for quick delivery of medically essential services. As we have seen in the above-mentioned work, any researchers are modeling and contributing to the development of varied types of trackers, ie. medicine tracker, ventilator tracker, Contact Tracing Tracker, Vaccination Tracker, etc. Each of the trackers will have its way of collecting, modeling, presenting the specific domain data, although each one is working for a single pandemic situation. This is a major gap in the dissemination of crucial information. Also, one has to rely on different trackers for varied information required during a single medical emergency [15]. Hence, A generic approach for the development and deployment of any pandemic tracking, monitoring, or broadcasting system is needed at this hour.

Considering the need for the hour in this era of the Covid pandemic, the dashboard was presented with that data available since the research was limited to articles and datasets revealed till May 2020. Since visualization and ready-to-use services are always a better approach and always appreciated by the users, so it provides a visual and statistical representation and maps locations for the same in the best possible way using the R programming language.

The developed prototype showcased in this paper consists of a problem concerning the unawareness of the Country's available Medical Facilities at times of pandemics like Flu, Cholera, Coronavirus, etc. The issues and main challenges that motivated the development of the proposed framework will be discussed in this section. As it is very clear that COVID-19 is creating very disaster effects globally with India, people are getting panicked, emotionally unsecured, depressed, and in the stage of confusion, unaware about facilities provided by the government, regarding health care infrastructure of the country, regarding reliable news sources, symptoms and its prevention with the cure [17]. Amid all these things, It is necessary to understand and be aware of the national medical facilities and their preparedness and response plans for various such pandemics. A window showcasing adequate health infrastructure of the country will not only provide necessary medical information to the country's people but also help dignitaries to look into the serious obligatory medical requirements of one's nation or region or any particular area. This will not only help in improving the medical infrastructure but also reduce psychological stress.

The work below is divided into 05 sections, section 03 describes a generic and enhanced architecture for rapid development of a variety of trackers, the next section, section

04 showcases a prototype tracker developed using proposed enhanced architecture. The conclusion and future work are present in the last section followed by

III. PROPOSED ARCHITECTURE

In this section, an extensive description of our proposed architecture is present. We have designed one 04-layer generic client-server architecture and further improved it by amalgaming different layers of this architecture. The enhanced 03-layer architecture is found to be more realistic and ready for quick tracker development in the future. The 04 layered frameworks are shown in Figure 01. It works in a client-server architectural pattern framework, supports the addition of multiple client objects or functionalities, and open hooks for future addition or deletions. This architecture shares computing resources and the server hosts deliver and manage most of the services to be consumed by the client. In this architecture the user will get access to the application based on the proposed architecture just by clicking the URL, no such credentials are required to use the framework. No Authentication protocol will make this framework handy, intelligible, and comprehensible hence we may expect heavy traffic of users. To distribute the task over the web UI component and prevent it from crashing and overloading, we have used a load balancer in between the web component and the user layer. It serves as the single point of contact for users also distributes incoming application traffic across multiple targets. This increases the availability of the system. Huge user traffic needs high-performance computing on voluminous data so to accomplish this task we have added a most important layer of high-performance Servers as with every user action we may require various input/output calculations from the database. The volume of a real-time dataset is unpredictable, so despite using a MySQL Warehouse, we are taking multidimensional cubicles supporting no SQL databases and consuming SaaS for storage purposes. An important note to look at here is that we have used a separate layer for Web UI and Output Servers. This architecture is efficient to deal with voluminous data and high user traffic and provides features like supercomputing, parallel processing, high Accessibility, and Streamline Data Processing. Due to high traffic and enormous computing our architecture may suffer from the performance hit due to required handshaking and acknowledgments between different layers.

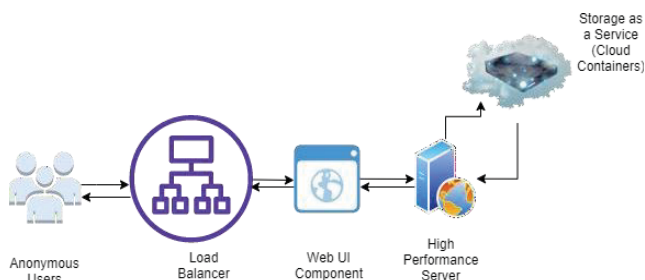


Fig. 1. Proposed generic 04 Layer Architecture for Pandemic Medical Tracker Development

Placing separate UI layer and server layer hit performance of our architecture due to required handshaking and acknowledgment between different layers. Loading the data for the first time on the UI from the server takes a bit more time also with client-side rendering we cannot predict the performance of the architecture. It also delays web crawling. It will also create inconsistency of data transfer between the

layer during poor or low connection. As data synchronization between these two layers is important and to ensure it, these layers take a lot of time to load data into the UI as the volume of data is very high.

To overcome this problem, we have modified our proposed Architecture wherein the WebUI and High-Performance Server are a part of a single layer. This enhanced architecture is given in Figure 02, has benefits like rapid development – as, in times of crisis, we may not have time to step up our framework and monitor each layer individually. This is generally termed Server-Side Rendering. The amalgamation of web UI layer and high-performance server refers to server-side rendering which is the ability of an application to exhibit the web UI on the server instead of rendering it in the browser [9]. Using this server sends the fully rendered page to the client.

The confined layer of server-based UI enables pages to load faster and give a better User experience. it also assists with loading the website when the user has a slow internet connection.[10] it plays a crucial role in Search Engine Optimization and correctly indexing web pages. Server-side rendering boosts user engagement by giving a fully rendered web page without any data loss. In our case the term of usage is high with a variety of user devices, it is clear that each device has a different configuration and Hardware model so it is better to send a fully rendered page to the user as per their browser requirement and request. as the volume of data is quite high hence necessary to handle the process of data visualization on the high-performance server itself so that the user gets the fully loaded web page with no wait.

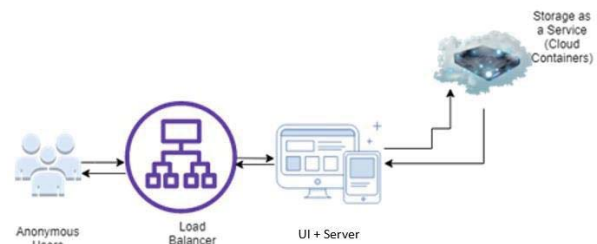


Fig. 2. Proposed advanced Layer 03 Architecture for Medical Tracker Development

Many languages are supporting this methodology. A few to mention are R, Python, Jackpack, etc. R provides an extensively usable package called Rshiny for server-side rendering. It's easy to use, learn and deploy in real-world application or tracker development. Apart from RShiny one can also use Bokeh which is a Python interactive visualization library that targets modern web browsers for presentation, spyre a server-driven UI web application Framework for providing simple modular UI for python, Jack Pack compose which is a new UI toolkit for native Android UI development that enables server-driven UI, in addition to these frameworks there are some other open-source libraries like epoxy from Airbnb, litho from Facebook proteus from Flipkart which can serve the need UI. server single layer is generally faster to make all the requests with the server than taking extra browser to server round trip. this will also improve infrastructure management and the handshaking system. this is search Engine Optimisation friendly, provide better performance as the code is shared with backend nodes, increase the efficiency of the application as the user machine is less busy, using a separate layer for UI and server will not only increase load

time but also delay web crawling which can be solved by combining UI and server [11].

IV. RESULTS

To understand the achievable functionalities, performance, and feasibility of the proposed architecture and its associated functionalities, we implemented a fully functional prototype. The primary objective of surveillance and monitoring during the pandemic is to track the course of the pandemic along with the country's medical facilities. Keeping this into view, we have used the proposed layer 03 architecture from the methodology section and developed a tracker application that can be directly used via URL by any internet-connected device. Any user can access the modules and functionalities available in the application without the need for credentials and authentication. Users need to provide location and GPS permission to use the application. For the development of this prototype tracker, we have taken a pandemic situation of COVID 19 and implied the same in the case of a developing country like India where there is an ample amount of population with limited medical resources for each one of the individual treatments. We enriched our system by adding other useful information like frequently asked questions related to the pandemic, helpline numbers provided by center or state, and user location-driven route locator system for finding the clearest route to a nearby hospital, labs, and medical practitioner using this architecture. The main strengths of using layer 03 architecture for the development of medical assistance tracker are the following:

It's a rapid application development architecture.

1. Enables to have a quick look at available medical facilities holistically or on any granular level, hence easy to make effective and more accurate decisions.
2. Information is easily available and accessible i.e. a click away.
3. Works across platforms like android, iOS, etc.
4. Supports Voluminous data with high-performance computation.
5. Low in latency and high in Throughput.
6. Higher usability as all the required information is present in a systematic format.
7. Discovering new patterns and requirement classification is easier

A. Prototype process flow

A typical process flow of the developed prototype is given in Figure 03. The abbreviations used below are DCL meaning Data Collection Layer, DPL meaning Data Processing Layer, and DRL meaning Data Reporting Layer. Each Layer comprises a specific set of activities and helps achieve overall tracker functionalities.

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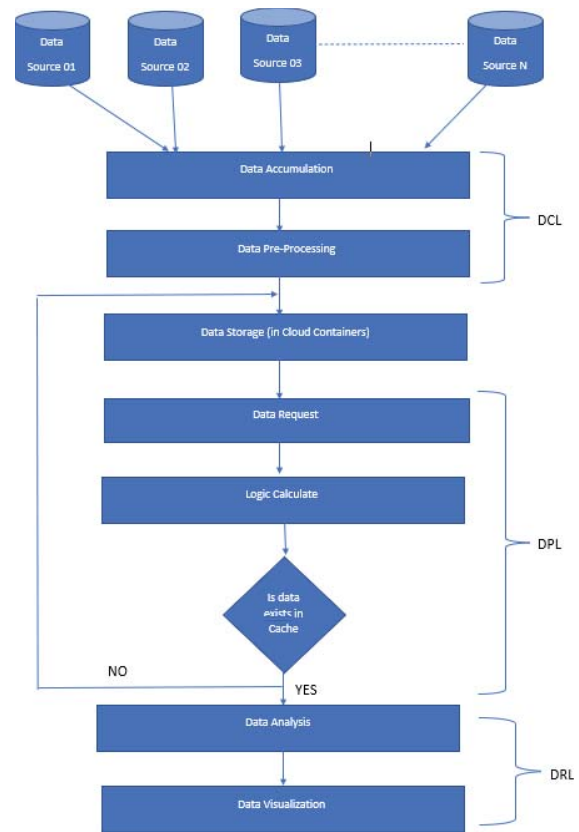


Fig. 3. Process Flow of Tracker based on Layer 03 Architecture

The first step is data collection from authentic data sources like global as well as government medical bodies such as the ministry of home and family affairs(MOHFW), AIIMS, Indian Council of Medical Research(ICMR), NHP (National Health Program), etc. The next step is data pre-processing which may include typical data formatting, cleaning, filtering, tuning, and finally merging this dataset. The logical division of data before actually sending it to cloud containers is also performed here. Here, we divided the dataset based on region as he has taken the tracker development for India, here the impact of this virus can be readily understood using zones or regions. This logical division is open for customization dependent upon the impression of the current pandemic. This division is helpful in quick interactive input-output operations. After a logical break-up, we are ready to send this data to our cloud containers. These cloud containers have virtual memory and therefore help scale our application in case of huge information. After saving the data, the role of a fancy client and powerful server core development. As we are using layer 03 architecture the next step here is to find a suitable server-side rendering language that comes into the picture. We have used the Rshiny package of language R for this purpose as it is easy to learn. Using SaaS along with Rshiny can help us in expediting our development work by leaving application bottlenecks like infrastructure management, performance management, handshaking management, layer interaction management, etc to the service provider's end. Also, the clarity and beautification of client-side web UI are not required to be done by the developer as varied styling components are already incorporated in the UI layer by the framework itself. As we were required interactive charts wherein a user can explore and filter tailor-made queries at the run time, this Rshiny along with SaaS gives no halt at the user's end. This package is cost-effective, performance effective and freely available, and helpful in rapid application development. After

setting up client and server interaction, we try to optimize our comprehensive framework performance. For this, we have used the server-side cache implicitly present in Rshiny. Before actual query triggering to cloud containers or storage, we check the same in our server cache. If the result is present we directly send the same to the client for rendering purposes, else the query is being sent to the cloud containers for fetching out the tailored result. The result is handed over to the server if there is any further calculation to be processed and then forwarded to the client-side for rendering purposes. Finally, data analysis and exploration analysis is done using visualization methods like bar, graphs, chart, etc.

B. Layers description

The process flow comprises three different layers starting from the accumulation of data and ending with its reporting. In between, there is a processing layer which is the core of this workflow. The request of a particular data is being in the processing layer and response from virtual data stores is received by this layer only. Let's discuss the roles and responsibilities of each layer along with their sublayer in the next section.

1) *DCL (Data Collection Layer)*: The heart of any data science-driven framework is data nevertheless the quality of the dataset should be way high. The data should not only be authentic, complete, consistent concerning time as well as content and acquisition of it should be specifically from reliable sources. Not only in this pandemic (Covid 19) situation but also, during any pandemic situation medical facilities hold utmost importance, and deploying this rapid solution for propagating the authentic information among all can be proven as revolutionist for reducing anxiety as well as for enhancing the knowledge about present medical facilities.

This layer is responsible for the collection of data from varied data sources. Here comes the role of Preprocessing activities like cleaning, formatting, transformation, aggregation, segregation, treating sparsity, removing, or swapping missing values, etc. It is recommended to perform outlier detection twice before feeding the tuned dataset to the processing layer. We have performed the same before and after the pre-processing methods. The main aim of the data collection layer was to pull together a list of trusted, valuable, statistically sound, and scientifically managed hospitals, testing centers, and help centers. By comprehending the subtleness and intricacy of the burning topic, systematic research was performed using "Covid-19", "Coronavirus dataset", testing labs and hospitals" keywords and later the data (links in references) was consumed as secondary quantitative data in the prototype development.

2) *DPL (Data Processing Layer)*: The refined data is now ready to be fed to Data Processing Layer. It comprises two sub-layers namely, Data Request Layer (Client) and Logic Calculation Layer (Server). Data Request Layer is typically for Client-Side containing Rich Web UI, Interactive forms to fetch the required data from High-end Calculation Servers. Logic Calculation Layer is responsible for fetching the query results from cloud containers as per client requests. Since we are using Rshiny as a server-side rendering language these two are part of a single class and can be used without suffering from client-server handshaking hang-ups. For optimization purposes, implicit server cache is being used

and queries are being checked before reaching storage containers. The result is directly returned by the local server in case the cache had that result previously-stored, else the request has been forwarded to the SaaS. In the case of our developed prototype the results for modules like frequently asked questions, a quick helpline is always present in the server cache, wherein other results like customized route locator, the user-specific query result is fetched from cloud containers. The load on page rendering is also gets decreased by this cache mechanism.

3) *DRL (Data Reporting Layer)*: On receiving the response from virtual stores, the next step is to visualize the same by using various EDA techniques like charting, graphical representation, interactive pies, etc. This layer is responsible for these mentioned activities along with the data analysis layer. This layer is responsible for meeting the end goal of any tracker i.e. dissemination of useful information from data generators to its users.

C. Ptototype description

The major modules of our developed prototype using architecture 02 are listed below:

1) *Dashboard*: The Dashboard or Home page will contain information about different zones in the country. This can be drilled down further to the region. The dashboard, in Fig 4 contains every detail about the number of hospitals, beds, medical emergency kits, available ventilators per hospital, testing labs, etc.

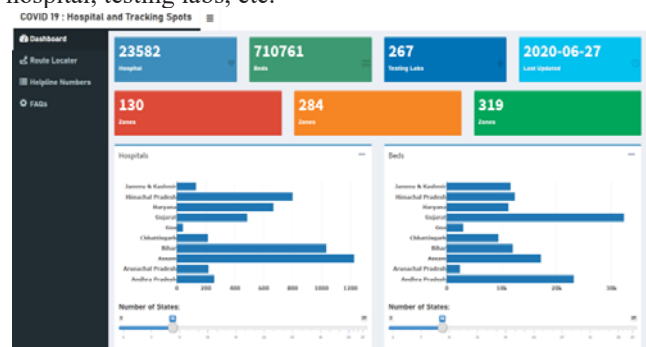


Fig. 4. Tracker Home Page

2) *Route Locator*: This showcases various routes from a user as a source to medical emergency destinations like nearby testing labs, pandemic dedicated hospitals, medical physical facilities, etc. A marker shows the user's live location and gives a connected route for all the locations of the nearby labs and hospitals in the network. And to get a detailed view of the locations one can easily zoom in and zoom out the map to steadily track the path as shown in Fig 5.

3) *A Quick Helpline*: This page comprises various contact numbers available in regions, district level, country-level, or globally broadcasted helpline details. The list of helpline nos. for the ambulance and hospitals in all the states and the union territories as generated by the Govt. of India is also made available for the public here. It enlists the helplines declared for essential service during the coronavirus lockdown and seeks help for those who are sick.

4) *Frequently Asked Question*: The main goal of any tracker is to omit the myths related to pandemic diseases. This

module can help the countrymen to get relief from the folk tales about health safety, precaution, detection, mitigation plan during any pandemic. This module will be able to provide the correct answers to general questions related to overall health care. It will also highlight the advisories to be followed by the citizens and basic knowledge-based questions and answers based on the pandemic causing the coronavirus (Covid-19). It can take into consideration the do's and don'ts to reduce the risk of catching the infectious virus or spreading it and most likely questions like that.

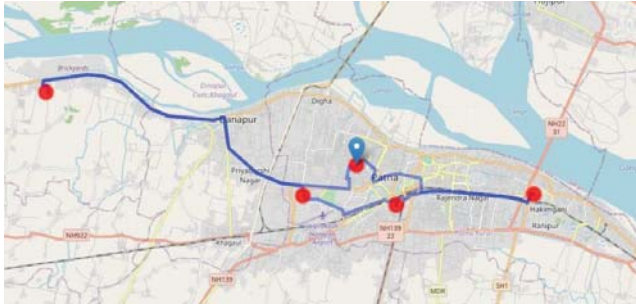


Fig. 5. Route Locator showing routes - Blue Lines for Path - Red Spots for Destination - Marker for your current location

5) *Chatbot*: This module contains an interactive AI-enabled chatbot. It helps users to ask questions anytime and try to provide answers as per stored data. In our work, the repository we have taken is frequently asked questions. If the questions match with the stored values, the respective answers get populated, if not, the question gets stored in the cloud container's question repository and answered at a later stage.

The proposed architecture supports the development of a ready-to-use internet-based application that can be accessed on any platform or device. The viability of this architecture is successfully showcased in the implementation section. Overall, the application developed using the proposed architecture is found to be efficient, interactive, and user-friendly.

V. CONCLUSION

Any crisis triggers an alarm to every country's current as well as future economy. The economy directly impacts the medical situation available as well as plans for prevention, mitigation, control, and emergence out of such an emergency. In the future, this architecture can be expanded depending upon the requirements of the situation keeping its core intact and providing a tailor-made solution at various states of affairs. As we are facing a new virus and disease, future directions and scenarios should be further assessed and added. The performance analysis of these two architectures can be benchmarked on different data loads near future. Further new functionalities like local or global medical strategy development, operational decision making, descriptive analysis, prescriptive analysis, medical planning, facility planning can be added using this architecture. Natural Language Processing can also be employed to enhance chatbot systems that can remotely communicate and provide consultation to people and patients about the Pandemic. AI can be used to eradicate fake news to ensure clear, responsible, and reliable information about pandemics such as evidence related to viruses, government social policies, and pandemic prevention and control guidelines.

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